

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PNG 322 — Past Questions & Materials
by Renance

Course: PNG 322

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — PNG 322 Well completion and System design 2023/2024

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
SECOND SEMESTER EXAMINATION, 2023/2024 SESSION

PNG 322: WELL COMPLETIONS AND SYSTEM DESIGN
TIME ALLOWED: 3HRS: 11/10/2024. UNIT: 3
ANSWER ANY FIVE QUESTIONS ONLY.

QUESTIONS.

- (1). a.(i) What is well Completion Operation? State and briefly explain its fundamental requirements and classification.
(ii) When does the Completion phase begin? (4 MARKS)
b. Briefly explain the function of the following completion components:
Packers, Nipples, Sliding Sleeves, Gas Lift Mandrels and Surface Controlled Subsurface Safety Valves (SCSSV). (4 MARKS)
c. Describe the completion categories listed below, state four advantages and three disadvantages for each.
(i) Cemented and Perforated liner/casing completions (ii) Screen/slotted liner completions. (iii) Open hole completions. (7 MARKS)
- (2) a. Briefly explain with the aid of a diagram what you understand by:
(i) Formation pressure. (ii) Overburden pressure. (iii) Formation foreshortening. (6 MARKS)
b. Given dexp vs. depth as shown in Figure 1 below. Estimate the formation pressure at 13,000 ft using Rehm and McClendon, and the Zamora correlation. (5 MARKS)

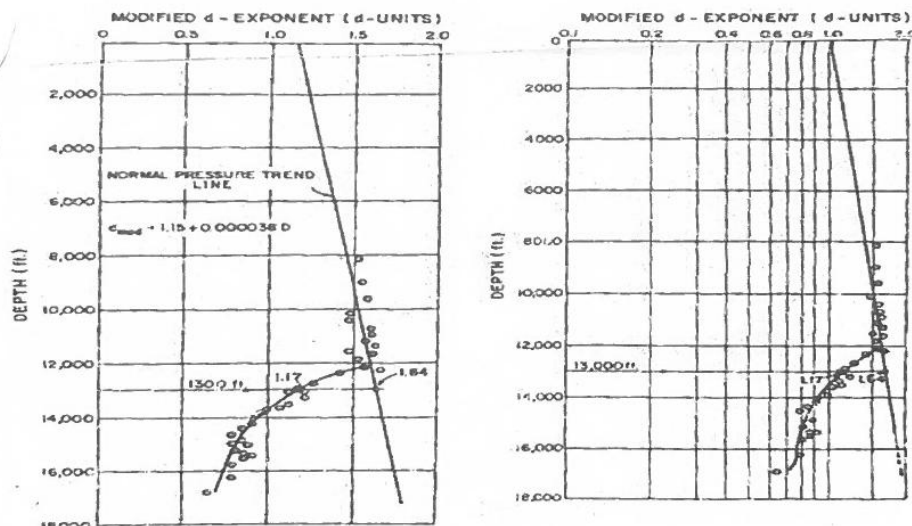


Fig 1

- c. (i). Identify and explain two ways gas cutting of mud may occur.
(ii). Consider a gas sand which was encountered in the U.S. Gulf Coast area. If the water-filled portion of the sand is pressured normally and the gas/water contact occurred at a depth of 5,000 ft, what mud weight would be required to drill through the top of the sand structure safely at a depth of 4,000 ft? Assume the gas has an average density of 0.76 lbm/gal. (4 MARKS)

p1 - down 0.052

3) a.(i) State and explain the phases of casing design.

(ii).What do you understand by the statement 'the detailed design of a casing string'?

(iii).Explain with the aid of a flow diagram, the process involved in the detailed design of a casing string.(8 MARKS)

b. An oil well in Ewreni town in the Niger delta region with 13-3/8" intermediate casing set at 11,750ft, having mud weight of 15ppg. Top of spacer is given to be 2,750ft with spacer weight of 14ppg. top of cement at 3000ft with cement weight 16ppg. Plug bump pressure is 3000psi.

2558

Calculate the net burst load at casing shoe. (7 MARKS)

(4) a Briefly explain the following:(i) Reasons of Cementing an oil well.(ii) Application of Directional Drilling.

(iii) Purposes of Installing Casing.(iv) Bumping plug.(v) Types of Casing. (6 MARKS)

4b(i) In a drilling programme done in the North-bank of the Niger Delta region, Calculate the total load of running a 15 3/8", 72 lb/ft intermediate casing to 10,750ft with inside diameter 13.347", mud weight 13ppg, instantaneous velocity 6ft/sec. (7 MARKS)

(ii). What is the purpose of carrying out a casing test. (2 MARKS)

5) a(i). State and Briefly explain five fundamental properties which must be considered when designing any cement slurry.

(ii).What is Dog leg severity?(ii)List any five cementing equipment and their functions. (8 MARKS).

(iii).The ROP is a function of many factors other than the formation type and formation pressure; state five additional factors.

b. FUPRE 007BIS well was finally drilled as designed, with 9 5/8" production casing set at 11,500 ft with mud weight 11 ppg, top of cement at 7,000 ft with 16 ppg cement weight. Spacer 5,000ft with 13 ppg spacer weight. Calculate the net collapse load at casing shoe and represent it graphically. (7MARKS)

(6) Determine the cement volume calculations of a 9 5/8" casing of a well to be cemented in place with a single stage cementing operation. The appropriate calculations are to be conducted prior to the operation.

given the details of the operation are as follows: 9 5/8" 47lb/ft casing set at: 12,800'. 12 1/4" hole at 13,810'

13 3/8" 68lb/ft" casing set at: 6200' above shoe. Calculate the total cement slurry, Amount of cement and mix water,

Amount of additives and the displacement volume.

Assume gauge hole, add 10% excess in open hole. If the casing is to be cemented with class G cement with the following additives 0.2% D13R (retarder), 1% D65 (friction reducer) slurry density 15.9 ppg. For nat pump 12-p-160, 7" liner 97% eff, 0.138 bbl/stk (15 MARKS). see fig 2 below

The casing is to be cemented with class G cement with the following additives:

0.2% D13R (retarder)

1 % D65 (friction reducer)

Slurry density

= 15.9 ppg

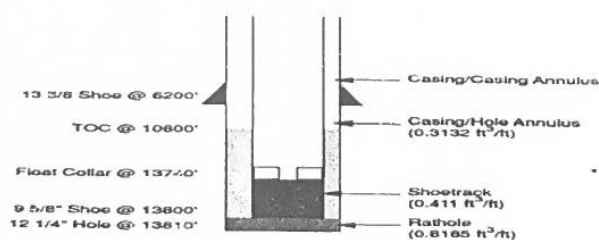


fig 2.

1.14
4.96
94